

Smart Horse

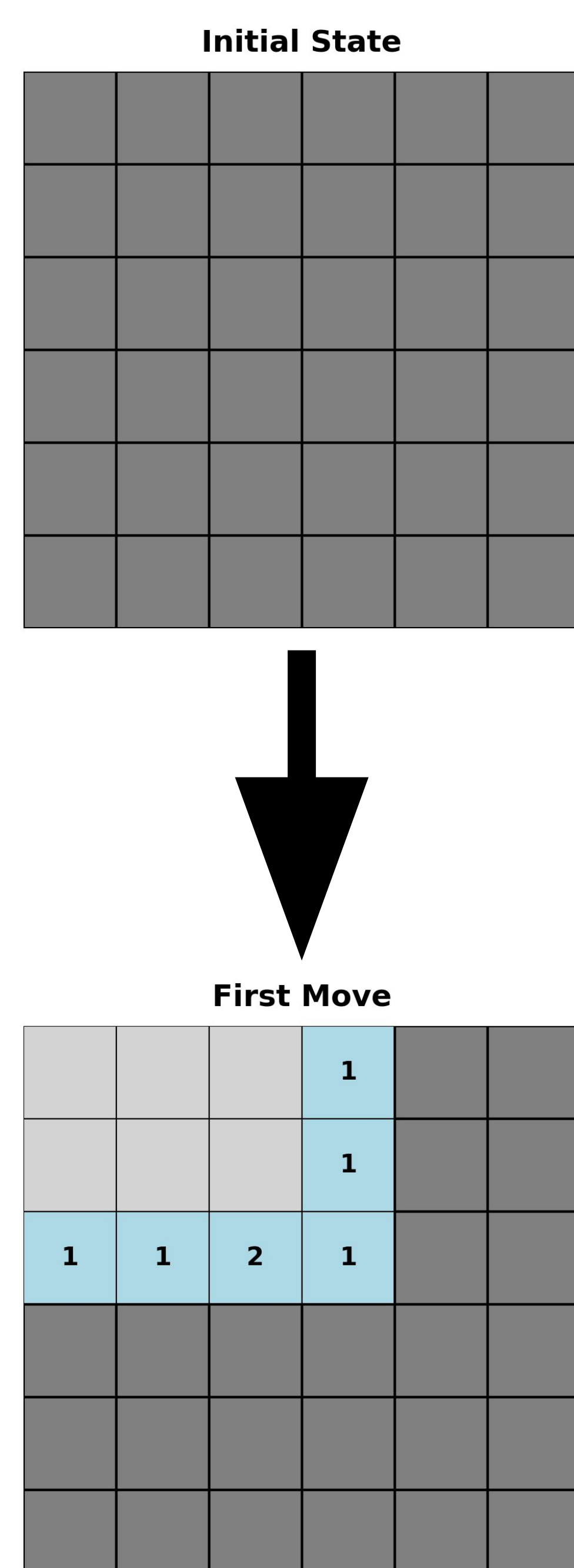
Self-Directed Reinforcement Learning

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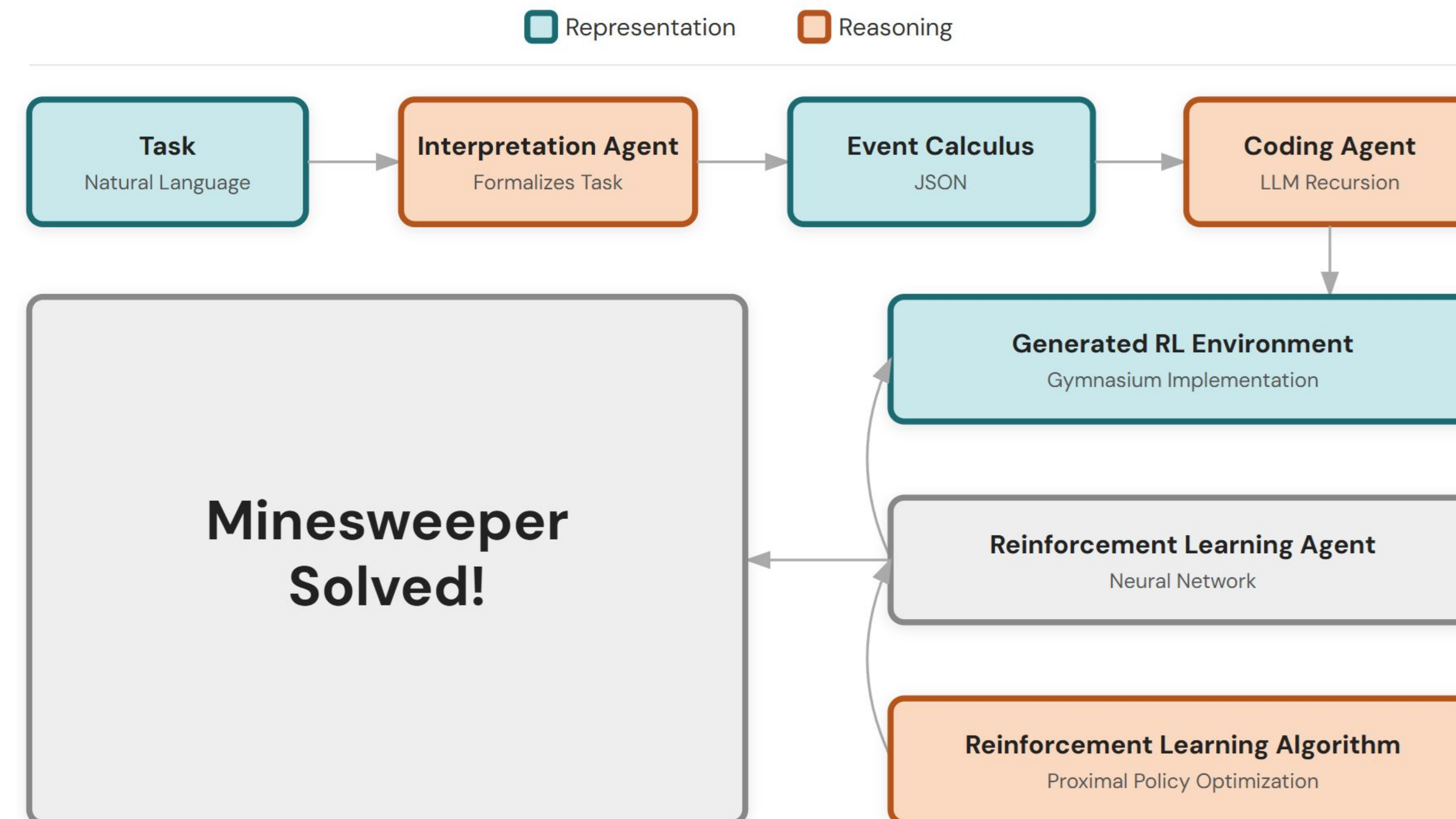
Introduction

Reinforcement learning requires bespoke environments made to solve particular problems. *Smart Horse* enables quick environment generation via natural language descriptions.

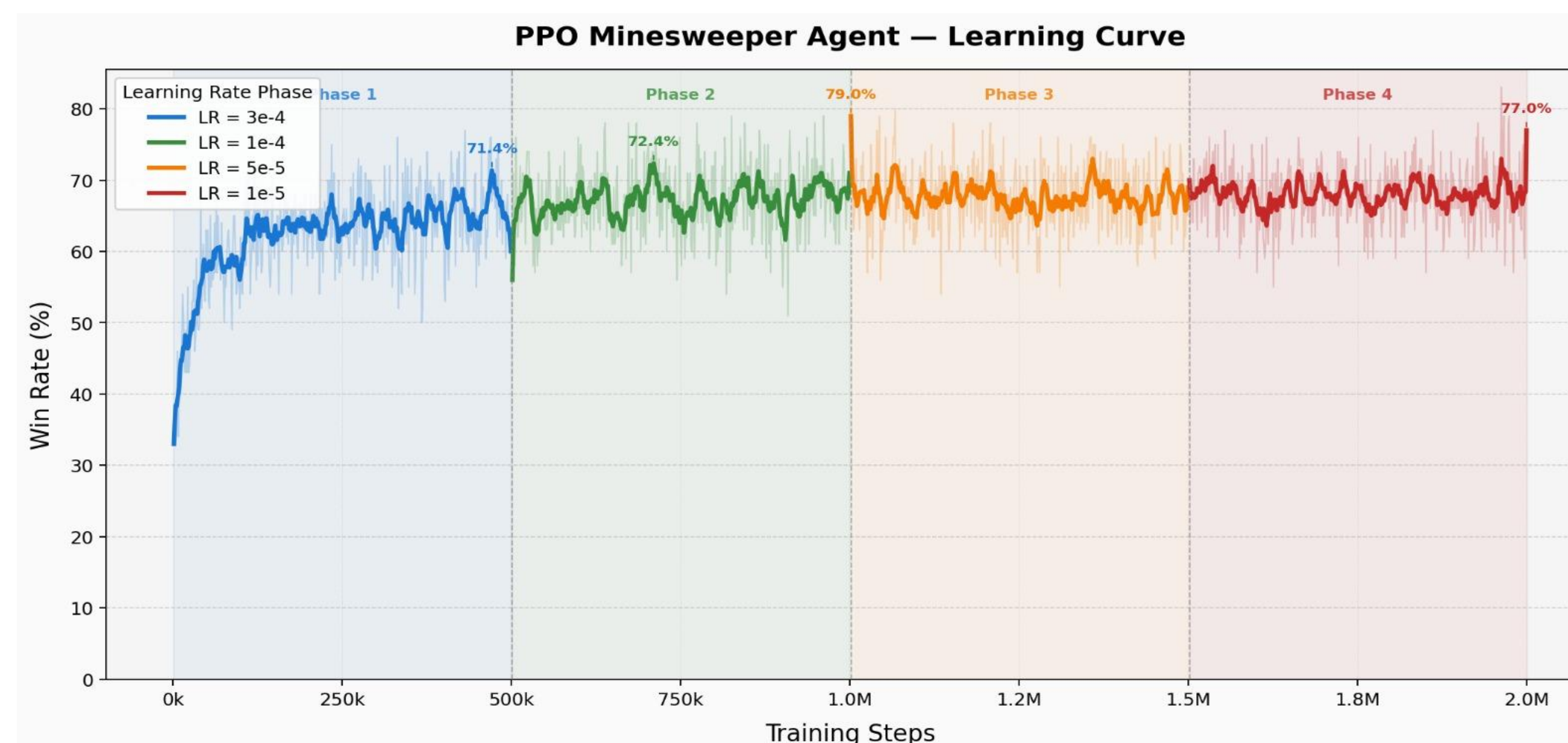
Minesweeper



Project Architecture



Phase	Learning Rate	Peak Win Rate	Final Win Rate	Mean Win Rate	Mean Mine Rate	Avg Episode Len	Avg Reward
Phase 1	LR = 3e-4	77.0%	60.0%	62.2%	37.8%	5.7	+32.740
Phase 2	LR = 1e-4	79.0%	71.0%	67.2%	32.8%	5.8	+33.956
Phase 3	LR = 5e-5	80.0%	69.0%	67.7%	32.3%	5.9	+34.165
Phase 4	LR = 1e-5	83.0%	77.0%	68.0%	32.0%	5.9	+34.310



Conclusions

The Smart Horse reasoning structure is capable of generating simple environments useful for reinforcement learning. The pure natural language input is essential because it opens the door for LLMs to access RL reasoning *at inference time*. The RL and NLP literature has shown that RL models can solve classes of problems which are difficult for LLMs alone, such as complex scheduling, spatial navigation, and high precision prediction.

Next steps:

- Automate Reinforcement Learning
- Agentic review of Reinforcement Learning learned strategies
- Expand to new use cases (Ex: checkers, tic tac toe, etc.)